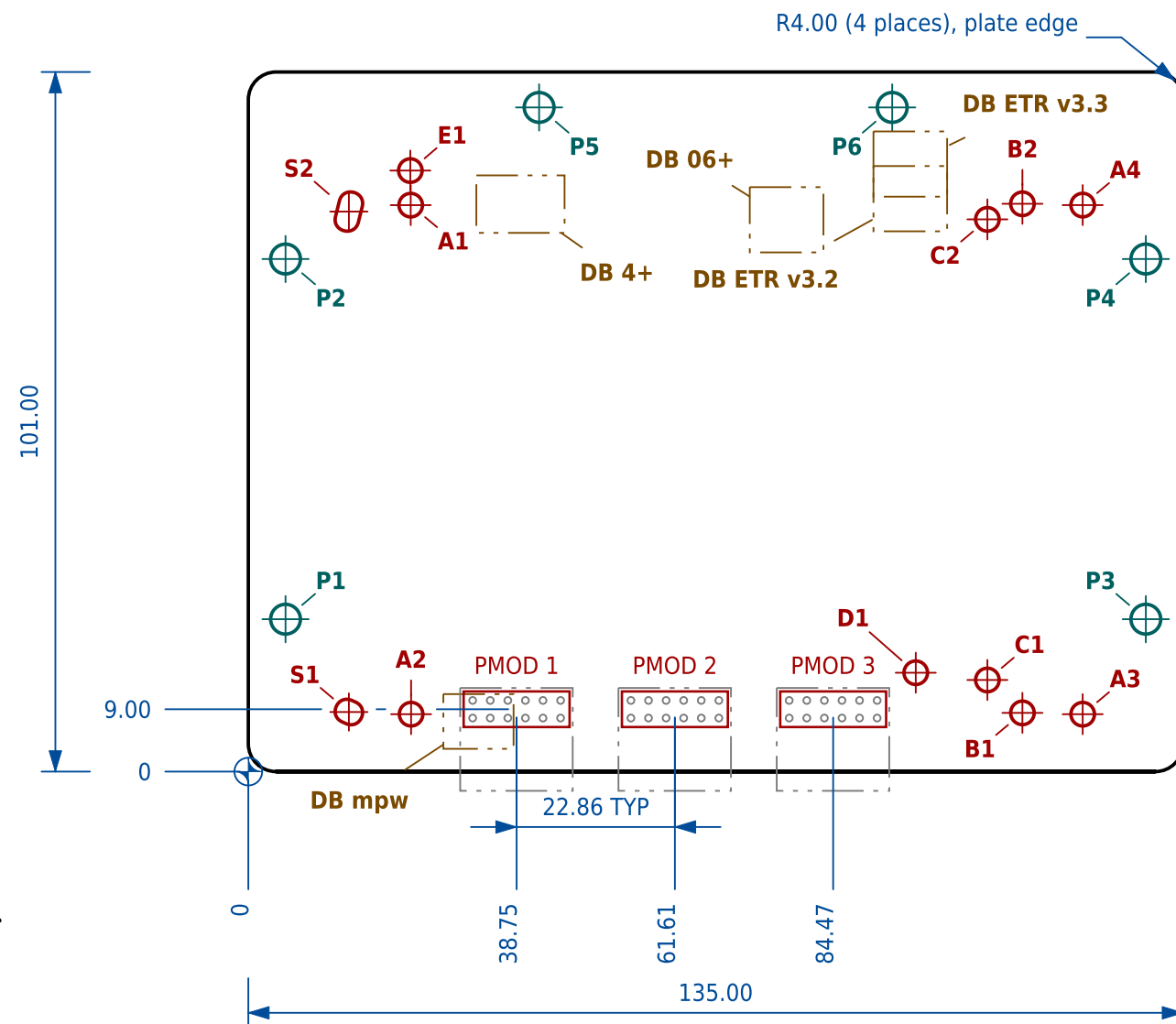




NOTES

1. All dimensions in millimetres. The datum symbol marks the origin: the plate's lower-left corner, X right, Y up, seen from the side the board mounts on.
2. Hole positions are tabulated, not dimensioned on the view: there are too many to dimension and keep it readable. A slot row gives its two end centres; DIA/WIDTH is the slot width, LENGTH is overall.
3. USED BY reads <group>:<hole>; MT1 to MT4 are the board's own hole IDs. Groups, by board revision: DB mpw = tt123-v2.2.5, tt123-v2.2.6; DB 4+ = v1.2.2, v1.2.3; DB 06+ = v2.0.1, v2.1.0, v2.1.2; DB ETR v3.2 = v3.2; DB ETR v3.3 = themselves, no shuttle yet.
4. A board hole's ID letter is the revision it is there for -- A DB mpw, B DB 4+, C DB 06+, D DB ETR v3.2, E DB ETR v3.3 -- numbered within it. S is a slot and P a plate fixing. Four features serve two revisions; each keeps one ID.
5. The three PMOD envelopes are not machined features. They mark where the Pmod host pin fields end up, the same place for every board revision. That is the point of the plate.
6. Fit the board on standoffs. A board uses only the holes and slots its USED BY row names.

NOTES (continued)

7. Board holes and slots are 3.4 mm, a close clearance fit on M3. Plate fixings are 4.3 mm, a clearance fit on M4. A slot spans two revisions whose holes are too close together to drill separately.
8. The Pmod host connector bodies overhang the plate's front edge by 2.78 mm, so a peripheral module plugs into clear air.
9. Cut file: tt-generic-mounting-plate.dxf, four layers. Sizes here are finished sizes; allow for your cutter's kerf.
10. Fit the plate to its chassis before the board: the fixings sit in the border, which a board overhangs. Least material between features is 1.59 mm, between A1 and E1.
11. Drawing TT-MP-02 shows which holes each revision uses.

SOURCES

1. Derived from: `tinytapeout/boards.py` - Board outlines, mounting holes and Pmod host positions for every Tiny Tapeout demo board revision, extracted from the upstream KiCad files.

BOARD MOUNTING HOLES

ID	X mm	Y mm	DIA/WIDTH mm	LENGTH mm	USED BY
A1	23.45	81.78	3.40	-	DB mpw:MT3, DB ETR v3.2:MT1
A2	23.51	8.29	3.40	-	DB mpw:MT1
A3	120.51	8.29	3.40	-	DB mpw:MT2
A4	120.51	81.78	3.40	-	DB mpw:MT4
B1	111.81	8.47	3.40	-	DB 4+:MT2
B2	111.81	81.97	3.40	-	DB 4+:MT4
C1	106.75	13.22	3.40	-	DB 06+:MT2
C2	106.75	79.72	3.40	-	DB 06+:MT4
D1	96.40	14.27	3.40	-	DB ETR v3.2:MT2, DB ETR v3.3:MT2
E1	23.40	86.77	3.40	-	DB ETR v3.3:MT1
S1	14.24 / 14.80	8.72 / 8.47	3.40	4.01	DB 4+:MT1, DB 06+:MT1
S2	14.24 / 14.80	79.72 / 81.97	3.40	5.72	DB 4+:MT3, DB 06+:MT3

PLATE FIXING HOLES

ID	X mm	Y mm	DIA mm
P1	5.37	22.00	4.30
P2	5.37	74.00	4.30
P3	129.63	22.00	4.30
P4	129.63	74.00	4.30
P5	42.00	95.89	4.30
P6	93.00	95.89	4.30

LEGEND

- | | |
|---|---|
|  | Plate edge |
|  | Hole or slot for a demo board fastener |
|  | Plate fixing, into the chassis |
|  | Slot axis |
|  | Pmod host pin field, where a board's hosts land |
|  | Pmod connector body, overhanging the front edge |
|  | USB-C connector, one position per board version |
|  | Dimension, extension and leader |

SOURCES (continued)

2. Pmod host pitch:
https://digilent.com/reference/_media/reference/pmod/pmod-interface-specification-1_2_0.pdf - Digilent mandate .90 in (22.86 mm) between adjacent host ports on a board edge, which is why one plate can serve every revision.

TT GENERIC MOUNTING PLATE

Accepts every demo board revision, Pmod hosts fixed in place

DRAWN generated	VERSION v0.0-90-g2879ddc+	UNITS mm	SCALE 1:1
SIZE A3	SHEET 1 OF 2	DRAWING NO TT-MP-01	REV A
MATERIAL 3 mm acrylic or 1.6 mm FR4		GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 allow for cutter kerf	